Definition (Stacking semicategory)
A *stacking semicategory* is a semicategory $\mathbf{C}$ with the following additional constituents and properties.

<u>Constituents</u>

- ▹ A stacking operation $\otimes : \mathrm{Ob}_{\mathbf{C}} \times \mathrm{Ob}_{\mathbf{C}} \to \mathrm{Ob}_{\mathbf{C}}$.
- ▹ A stacking operation $\otimes : \mathrm{Mor}_{\mathbf{C}} \times \mathrm{Mor}_{\mathbf{C}} \to \mathrm{Mor}_{\mathbf{C}}$.

<u>Conditions</u>

- ▹ The two operations $\otimes$ and $\otimes$ are compatible in the sense that

$$\frac{f_1 : X_1 \to Y_1 \qquad f_2 : X_2 \to Y_2}{f_1 \otimes f_2 : X_1 \otimes X_2 \to Y_1 \otimes Y_2}$$